

RAFLY ANUGRAH SYAHPUTRA

Data Scientist | Machine Learning | Applied AI

South Jakarta, Indonesia | raflyanugrahsyahputra.1@gmail.com | +62 813-3070-4109
linkedin.com/in/rafly-anugrah-syahputra | github.com/Conquer5

PROFESSIONAL SUMMARY

Data Science undergraduate (GPA 3.67/4.00) with hands-on experience in machine learning, data pipelines, quantitative analysis, and applied AI systems. Built a 60-minute CNN-LSTM rainfall nowcasting model at BMKG using AWS observations and Himawari-9 imagery, achieving 95% accuracy, 0.88 F1-score on significant weather events, and CSI 0.790. Also developed operational data workflows at PLN and production-oriented AI projects focused on grounded outputs, evaluation, reproducibility, and reliable deployment.

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, JavaScript, Pandas, SQLAlchemy, PostgreSQL/pgvector, MySQL, SQLite

Machine Learning & AI: TensorFlow, Scikit-learn, NLP, CNN/LSTM, Statistical Modeling, Time Series, Gemini API, LangGraph, LangChain, RAG, LLM Agents

Data Engineering & Automation: ETL/ELT, Apache Airflow, Celery/RabbitMQ, Asyncio, Data Validation, Playwright, Selenium, Web Scraping

Analytics & MLOps: Power BI, Tableau, Looker Studio, Streamlit, Backtesting, Docker, Linux/systemd, Git/GitHub Actions, Pytest

PROFESSIONAL EXPERIENCE

Data & Automation Intern | PT PLN (Persero), Pusdiklat

Apr 2026 - Aug 2026

- Built an ETL workflow that synchronized operational KPI data, processed 50+ Excel workbooks from 23 training units with Pandas/openpyxl, validated records, and stored clean outputs in SQLite for reporting.
- Developed KonMan, a five-page Streamlit dashboard for unit performance monitoring, data-quality checks, and automated PDF/Excel reporting.
- Created Looker Studio and Google Apps Script automations, plus a Selenium proof of concept for extracting Nota Dinas PDF data into outgoing-letter web forms; prepared Docker, systemd, and Synology handover guides.

Data Science Intern | BMKG, Juanda Meteorological Station

Jun 2025 - Dec 2025

- Automated a 24/7 Python ingestion workflow that replaced manual entry every 30 minutes and reduced operational workload by an estimated 80+ hours per month.
- Developed a CNN-LSTM model for 60-minute rainfall nowcasting using AWS observations and Himawari-9 satellite imagery, reaching 95% accuracy, 0.88 F1-score on significant weather events, and CSI 0.790 while outperforming baseline GRU models.
- Deployed real-time model outputs through an interactive Streamlit dashboard for operational review by forecasters.

Founder & Freelance Data Consultant | YoData

Oct 2024 - Present

- Completed 40+ data projects across machine learning, statistical analysis, data cleaning, and dashboard development for research and small-business use cases.
- Built NLP classification workflows using BERT, Random Forest, and TF-IDF, and delivered analysis in Python, R, SPSS, and EViews.
- Designed interactive dashboards in Power BI, Tableau, and Looker Studio for client reporting and decision support.

SELECTED DATA SCIENCE & AI PROJECTS

StockPulse AI | Financial Intelligence & Quantitative Decision-Support System | [Live Showcase](#)

Python, Gemini API, Pydantic, SQLite, JavaScript, REST API, Systemd

- Architected a hybrid AI and deterministic decision pipeline for the Indonesia Stock Exchange, combining LLM-based news fact extraction with quantitative rules to reduce hallucinated financial reasoning.
- Engineered multi-source ingestion from 8+ tier-classified sources, official IDX foreign-flow data, and broker summaries with heuristic deduplication and date-conflict quarantine.
- Built an immutable prediction ledger and 5-layer evaluation framework for grounding, decision consistency, and H+5/H+20 forward outcomes with zero lookahead bias, realistic fees/slippage, and ATR-based risk levels.

Autonomous AI Research Radar | Evidence-First Autonomous Research Agent

Python 3.12, Gemini, Hermes Agent Runtime, Pydantic v2, SQLite, Asyncio, Tenacity, Systemd, Pytest

- Architected an evidence-first research agent that separates Telegram UX, orchestration, and tool execution through authenticated HTTP boundaries, improving traceability and grounding.
- Built multi-source retrieval across arXiv, GitHub, Brave Search, and RSS with recency/velocity scoring, context-budgeted chunking, and SQLite deduplication.
- Implemented a copy-resistant consensus engine and a 32-case adversarial benchmark across 7 AI domains, measuring 100% grounding rate, citation coverage, and contradiction detection.

AI Career Agent | Human-in-the-Loop Career Intelligence System

Python, LangGraph, LangChain, Gemini, Playwright, SQLAlchemy, SQLite, Pytest

- Architected a stateful career research workflow for multi-platform job discovery, vacancy scoring, and candidate-job matching using LangGraph and Gemini.
- Designed a grounded form-filling engine that evaluates dynamic application fields only against verified CV/profile facts, with human checkpoints for salary, visa, CAPTCHA, and other sensitive inputs.
- Implemented multi-tier Gemini routing, automatic rate-limit failover, 30-day response caching, and persistent workflow state to improve reliability and control API usage.

PersonaLab | AI-Powered Developer Storytelling & Branding Platform

Next.js, TypeScript, Gemini API, GitHub API, LinkedIn REST API, Server Actions

- Built an end-to-end platform that transforms GitHub commits, code diffs, and repository READMEs into factual, publication-ready LinkedIn updates.
- Implemented a 4-stage AI pipeline (Content Scout, Evidence Judge, Draft Writer, Final Reviewer) and cross-repository fact checks to prevent fabricated metrics or business-impact claims.
- Added repository-specific project memory and deterministic commit-SHA hashing to avoid repetitive narratives and redundant LLM calls, with GitHub extraction and LinkedIn publishing integration.

EDUCATION & RECOGNITION

B.Sc. Data Science | State University of Surabaya (UNESA)

Aug 2023 - Present

GPA 3.67/4.00 | 122 credits

- Finalist & Co-Author, SENADA 2025 National Seminar - distributed ETL pipeline for clickbait and news classification using Celery and RabbitMQ.
- Semifinalist, WordPress Case Competition 2024 - developed a data-driven digital talent strategy for Society 5.0.
- Finalist, FMIPA Ambassador Selection 2024-2025; awarded the title of Putra Konservasi.

LEADERSHIP & ACTIVITIES

Commission A Staff | Student Representative Council (DPM) FMIPA UNESA

Feb 2025 - Jan 2026

- Monitored student-executive work programs, evaluated organizational performance, and served as MC/moderator for faculty forums and seminars.

Official Faculty Ambassador | FMIPA Ambassador Association

Sep 2024 - Sep 2025

- Represented the faculty at university events and coordinated cross-faculty communication, event logistics, and community initiatives.

LANGUAGES

Indonesian (Native) | English (Intermediate)